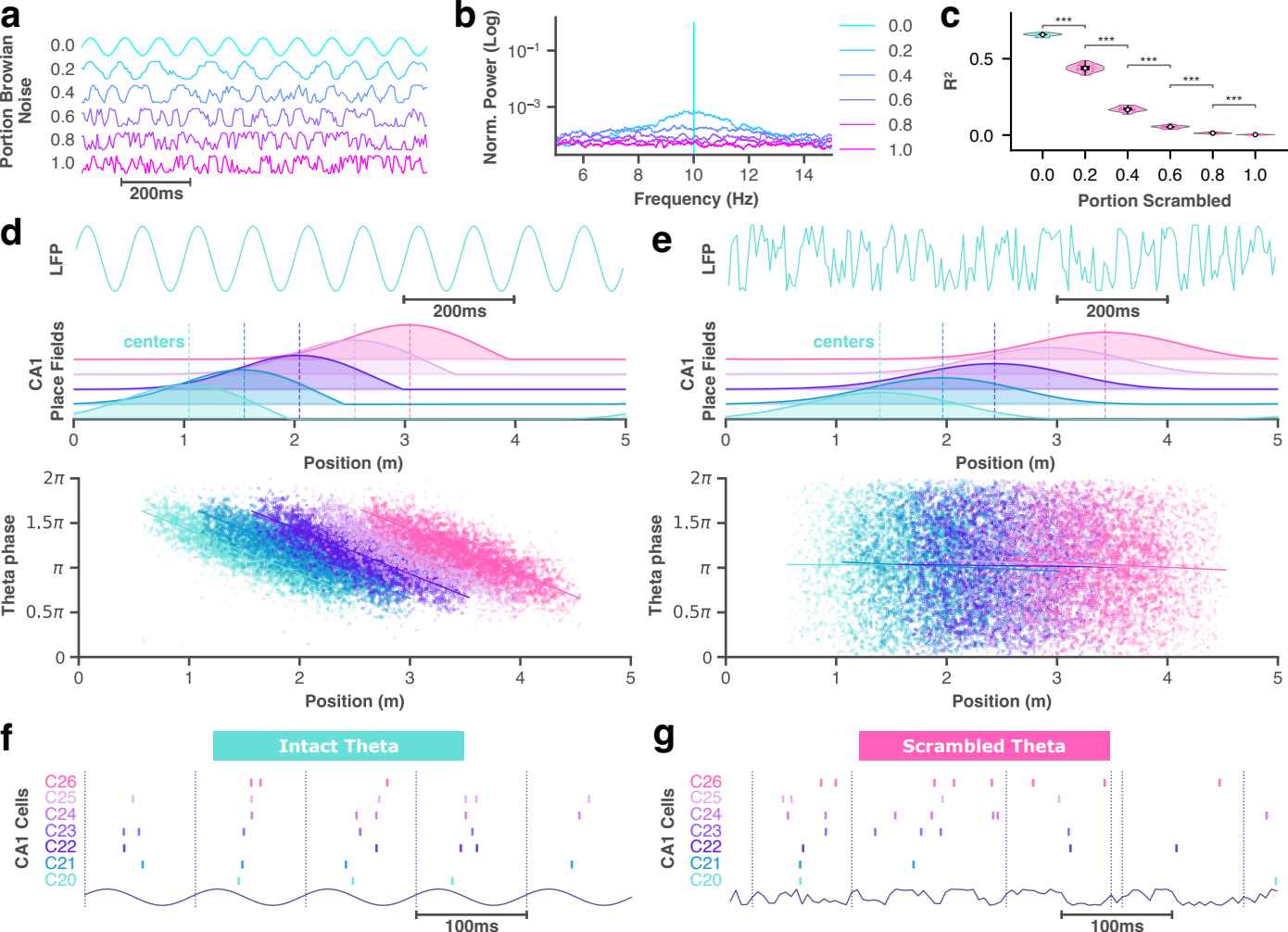

a
Portion Browian Noise
0.0
0.2
0.4
0.6
0.8
1.0
200ms
b
Norm. Power (Log)
10^{-1}
10^{-3}
6
8
10
12
14
Frequency (Hz)
0.0
0.2
0.4
0.6
0.8
1.0
c
R^2
0.5
0.0
***
***
***
***
***
0.0
0.2
0.4
0.6
0.8
1.0
Portion Scrambled
d
LFP
200ms
CA1 Place Fields
centers
0
1
2
3
4
5
Position (m)
Theta phase
2π
1.5π
π
0.5π
0
0
1
2
3
4
5
Position (m)
e
LFP
200ms
CA1 Place Fields
centers
0
1
2
3
4
5
Position (m)
Theta phase
2π
1.5π
π
0.5π
0
0
1
2
3
4
5
Position (m)
f
Intact Theta
CA1 Cells
C26
C25
C24
C23
C22
C21
C20
100ms
g
Scrambled Theta
CA1 Cells
C26
C25
C24
C23
C22
C21
C20
100ms